

ZoneSync Database System Case Study

ZoneSync has a number of Owners; each owner has a unique Owner_ID, Name, Email, and Password.

Each Owner can own many Zones, and each Zone must be owned by exactly one Owner. Each Zone has a unique Zone_Number, GIS_Location, and a predicted harvest date (Predicted_Harvest_Date) which is calculated based on growth factors.

Each Zone is planted with one Crop at a time, and a Crop can be grown in many different zones. Each crop has a Crop_Name, Planting_Date, and Growth_Stage.

ZoneSync also stores information about Farmers who help in managing the lands. Each Farmer has a unique Farmer_SSN, Farmer_Name, and may have multiple phone numbers (Phones). An Owner may employ many Farmers, and a Farmer can work for many Owners. Also, a Farmer can be assigned to manage multiple Zones, and each Zone can be managed by many Farmers.

The system monitors each Zone through Sensor Readings. Each reading is associated with one Zone and is identified by a Reading_ID. These readings include Moisture_Level, Temperature, pH_Level, and a set of soil nutrients (NPK_Levels) which include Nitrogen, Phosphorus, and Potassium levels.

Various Activities (such as Irrigation or Fertilizing) are performed in each Zone. Each activity has an Activity_ID, Activity_Type, Description, Amount_Consumed, and a flag to indicate if it was an automated process (Is_Automated).

An Owner may perform many activities, and a Farmer may also execute many activities within a Zone.

The system includes an AI Diagnosis feature for each Zone. Each diagnosis has a Diagnosis_ID, Image_Path, and Result. The AI Diagnosis analyzes the data provided by the Sensor Readings to detect plant health issues.

Whenever a problem is detected by the AI or the Owner, a Ticket is generated for the Zone. Each ticket has a unique Ticket_ID, Severity, Description, and Status (e.g., Open or Resolved). A Ticket can be triggered by a specific AI Diagnosis session, and the Owner is responsible for managing these tickets.